

ANDRÉS GÓMEZ RAMÍREZ

Cybersecurity Engineer & Security Researcher | PhD, Cybersecurity

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PROFILE

Cybersecurity engineer and security researcher with 15+ years of experience across penetration testing, vulnerability research, intrusion detection, and applied AI for security monitoring. PhD in Cybersecurity (Goethe University Frankfurt, in collaboration with CERN). Published researcher with 200+ peer-reviewed papers and 6 assigned CVEs, including a Microsoft-acknowledged memory-corruption vulnerability. Experienced across the full security lifecycle: ethical hacking and source code audits for banks and ISPs, large-scale distributed systems security at CERN, and founding a security startup. Comfortable working close to the metal (exploit development, fuzzing) and at the systems level (monitoring, infrastructure, DevSecOps).

CORE SKILLS

Offensive Security	Penetration testing, source code auditing, fuzzing, vulnerability research & exploit development, ethical hacking
Defensive Security	Intrusion detection & prevention, security monitoring for distributed/grid systems, distributed digital identity
AI / ML for Security	Deep learning, machine learning, NLP — TensorFlow, Keras, scikit-learn, pandas, NumPy, SciPy, Gensim
Languages	Python, JavaScript/Node.js, C/C++, Java, Solidity
DevOps / Infra	Docker, Ansible, Puppet, Foreman, Vagrant, Git, OpenStack, Linux (Debian, Ubuntu, CentOS, Tails)
Blockchain	Ethereum, Bitcoin, Polygon, Antelope (EOSIO), Dash, Nym — validator/node infrastructure, zero-knowledge proofs
Spoken Languages	Spanish (Native), English (Fluent), German (Basic)

PROFESSIONAL EXPERIENCE

Security Researcher — *Fluid Attacks*

2011 – 2013

- Performed penetration tests, source code audits, and vulnerability assessments for banks, software development companies, and ISPs.
- Conducted ethical hacking engagements; identified and reported multiple CVEs, including a Microsoft DirectShow memory-overwrite vulnerability (MS13-056).

PhD Researcher, Computer Science (Cybersecurity) — *Goethe University Frankfurt / CERN*

2013 – 2019

- Researched deep learning (generative models) and isolation-based methods for intrusion detection and prevention on the Worldwide LHC Computing Grid (WLCG), one of the largest distributed computing infrastructures in the world.
- Built a Python/Perl framework implementing the research; co-authored 3 peer-reviewed papers from this work.

DevOps Engineer — *Goethe University Frankfurt / CERN*

2013 – 2020

- Built and maintained the security monitoring system for the ALICE High-Level Trigger (HLT) computing cluster using Zabbix.
- Managed infrastructure-as-code and configuration management for the ALICE HLT cluster using Puppet.

Co-Founder & CEO — *Sakundi ÖU*

2020 – Present

- Founded and lead a security, privacy, and digital identity startup, setting technical direction and client strategy.
- Deliver security audits, monitoring solutions, and identity infrastructure to enterprise and protocol clients.

Security Developer — *Modulo Labs*

2025 – 2026

- Analyzed and developed cryptographic code and smart contracts for Panther, a privacy-enhancing protocol, with a focus on security review.

Part-Time Professor, Security/Blockchain — *Costa Rica Institute of Technology (TEC)*

2026 – Present

- Teach within the Cybersecurity Master's program, covering blockchain security fundamentals.

SELECTED PROJECTS

Tikuna: AI-based security monitoring system for the Ethereum P2P network (nodes & validators), funded by an Ethereum Foundation Academic Research grant. github.com/sakundi/tikuna

Arhuaco: PhD thesis project at CERN: deep learning and isolation-based intrusion detection for distributed high-throughput computing. github.com/kuronosec/arhuaco

Zikuani: Privacy-preserving identity verification system using zero-knowledge proofs. Backed by ZK Bankai and Polygon; DIF 2025 hackathon winner. github.com/kuronosec/zikuani

ZK Voto Digital: Secure voting system using Zikuani to verify voter eligibility (humanity, citizenship, age) without exposing personal identity. github.com/kuronosec/zk-voto-digital

EDUCATION

PhD, Computer Science (Cybersecurity) — *Goethe University, Frankfurt am Main, Germany*

2013 – 2019

Thesis: "Deep Learning and Isolation Based Security for Intrusion Detection and Prevention in Grid Computing"

BTA Certified Blockchain Security Professional — *Blockchain Training Alliance*

2021

BSc, Computer Science — *Universidad de Antioquia, Medellín, Colombia*

2003 – 2011

Awarded thesis: "Fast Selection of Radial Basis Function Network Centers Based on Marginal Data Analysis"

PUBLICATIONS & DISCLOSURES

- Gómez Ramírez, A., Al Sardy, L., Gómez Ramírez, F. — “Tikuna: An Ethereum Blockchain Network Security Monitoring System.” ISPEC 2023 Conference Proceedings.
- Gómez Ramírez, A., Lara, C., Betev, L., Bilanovic, D., Kebschull, U. (ALICE Collaboration) — “Arhuaco: Deep Learning and Isolation Based Security for Distributed High-Throughput Computing.” Journal of Grid Computing.
- Gómez Ramírez, A., Lara, C., Kebschull, U. (ALICE Collaboration) — “Intrusion Prevention and Detection in Grid Computing — The ALICE Case.” Journal of Physics: Conference Series, 664(6), 2015.
- Gómez Ramírez, A., Martínez Pedreira, M., Grigoras, C., Betev, L., Lara, C., Kebschull, U. (ALICE Collaboration) — “A Security Monitoring Framework for Virtualization Based HEP Infrastructures.” Journal of Physics: Conference Series, 898(10), 2017.
- 6 CVEs assigned: CVE-2011-4620, CVE-2012-1189, CVE-2012-2090, CVE-2012-2091, CVE-2012-4552, CVE-2012-5576. Including MS13-056 (DirectShow memory-overwrite vulnerability), published in the National Vulnerability Database.

REFERENCES

Available upon request, including former supervisors at CERN and Fluid Attacks.